

Chapter 8 — Codebooks, provenance, and notes

Dictionaries name variables; reuse also needs coded categories, source trails, and caveats

Learning objectives

- Write annotation notes that expose sampling limits or known data-quality issues

Codebooks for encoded categories

- A codebook maps numeric or abbreviated codes to full categorical meanings
- Healthcare, government
- Without a codebook, those codes are opaque and analyses risk silent mislabeling of classes

Example 8.9 — Medical Condition Codebook

- Example 8.9 — hands-on module
- Example 8.9 maps disease codes to readable names
- Explore the chapter example module
- View files: ``modules/chapter8/example9/``

Example 8.9 — listing

```
Code,Disease Name
```

```
1,Hypertension
```

```
2,Diabetes
```

```
3,Asthma
```

```
4,Chronic Obstructive Pulmonary Disease (COPD)
```

Provenance as data lineage

- Also called data lineage, records where data came from and what operations followed
- Links to other datasets that were merged or joined
- Provenance supports integrity checks and explains how validity may have changed

Example 8.10 — Retail Transaction Provenance Trail

- Example 8.10 — hands-on module
- Example 8.10 summarizes a retail trail
- That short narrative lets auditors reconstruct the processing path
- Explore the chapter example module
- View files: ``modules/chapter8/example10/``

Annotations and qualitative notes

- Annotations capture context that structured metadata cannot easily hold
- They warn users about sampling bias

Example 8.11 — Urban-Only Sampling Note

- Example 8.11 — hands-on module
- Example 8.11 records a critical sampling limit
- Without that note, downstream users might treat results as nationally generalizable
- Explore the chapter example module
- View files: ``modules/chapter8/example11/``

Takeaways

- Codebooks unlock encoded categories
- Together with metadata and dictionaries from the previous part
- Omissions in any layer create silent failure modes

Next

- Complete the quiz for this part
- The next part turns components into practice